3.3 prime residue classes and Fermat-Euler theorem (Def 3.13 — Cor 3.32)
3.4 cyclic groups (Def 3.33 — Thm 3.49)
from: script of Heidelberg University fall 2024 "Elementary Number Theory" lecture
[https://github.com/Hermann-SW/graphviz_latex_pipeline]

Rem 3.34 Every cyclic group is abelian.

Def 3.33 Group G is cyclic:
$\exists g \in G$ generator with
$G = \{g^n : n \in \mathbb{Z}\} =: < g >$

Ex 3.35 (a) group $(\mathbb{Z}/5\mathbb{Z})^\times = \{\bar{1}, \bar{2}, \bar{3}, \bar{4}\}$ is cyclic
(b) group $(\mathbb{Z}/8\mathbb{Z})^\times = \{\bar{1}, \bar{3}, \bar{5}, \bar{7}\}$ is not cyclic
(c) additive group $\mathbb{Z}$ is cyclic
(d) additive group $\mathbb{Z}/m\mathbb{Z} = \{\bar{0}, \bar{1}, \ldots, \overline{m-1}\}$ is cyclic

Ex 3.45 K field implies
$det : GL_n(K) \to K^\times$ homomorphism

Def 3.44 homomorphism
isomorphism, $\cong$

Ex 3.48 $(\mathbb{Z}/5\mathbb{Z})^\times \cong \mathbb{Z}/4\mathbb{Z}$
$\psi : \begin{cases} \mathbb{Z}/4\mathbb{Z} & \to (\mathbb{Z}/5\mathbb{Z})^\times \\ a & \mapsto \bar{2}^a \end{cases}$

Prop 3.46 G, H groups, $\psi : G \to H$ homomorphism:
(a) ψ injective $\iff \ker(\psi) := \{g \in G : \psi(g) = 1\} = \{1\}$
(b) ψ isomorphism $\iff \psi^{-1} : H \to G$ isomorphism

Thm 3.47 For cyclic group G:
$G \cong \begin{cases} \mathbb{Z} & \text{for } G \text{ infinite,} \\ \mathbb{Z}/|G|\mathbb{Z} & \text{for } G \text{ finite.} \end{cases}$

Ex 3.42 (a) $exp((\mathbb{Z}/5\mathbb{Z})^\times) = 4$
(b) $exp((\mathbb{Z}/8\mathbb{Z})^\times) = 2$

Def 3.41 G finite abelian group. Exponent
$\exp(G) := \min\{n \in \mathbb{N} : g^n = 1 \ \forall g \in G\}$

Prop 3.43 G finite abelian group:
(a) $\exp(G) \mid |G|$
(b) $\exp(G) = lcm(\{ord(g) : g \in G\})$

Prop 3.39 G finite abelian group,
$g \in G, m \in \mathbb{Z} : ord(g) \mid m \iff g^m = 1$

Prop 3.38 G finite abelian group:
G is cyclic $\iff \exists g \in G$ with $ord(g) = |G|$

Thm 3.49 G finite abelian group:
G is cyclic $\iff \exp(G) = |G|$

Ex 3.37 For $G = (\mathbb{Z}/5\mathbb{Z})^\times$:
(a) $ord(\bar{2}) = 4$
(b) $ord(\bar{4}) = 2$

Def 3.36 G finite abelian group.
Order $ord(G) := \min\{n \in \mathbb{N} : g^n = 1\}$

Cor 3.40 G finite abelian group.
$\forall g \in G : ord(g) \mid |G|$

Prop 3.29 G finite abelian group.
$\forall g \in G : g^{|G|} = 1$

Ex 3.31 $3^{19} \equiv 10 \ (mod\ 17)$

Thm 3.30 (Fermat-Euler theorem)
$n \in \mathbb{N}$. $\forall \bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times : \ \bar{a}^{\phi(n)} = \bar{1}$

Cor 3.32 (Fermat's little theorem) p prime:
(a) $\forall \bar{a} \in \mathbb{F}_p^\times : \bar{a}^{p-1} = \bar{1}$
(b) $\forall \bar{a} \in \mathbb{F}_p : \bar{a}^p = \bar{a}$

Ex 3.28 $|(\mathbb{Z}/n\mathbb{Z})^\times| = \phi(n)$

Def 3.27 Order $|G|$ of group
is number of its elements

Ex 3.24 Solutions for
$3x \equiv 1(mod\ 37) : L = 25 + 37\mathbb{Z}$

Prop 3.22 $a \in \mathbb{N}, \ \bar{a} \in \mathbb{Z}/n\mathbb{Z}$:
$\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times \iff \gcd(a, n) = 1$

Cor 3.23 $n \in \mathbb{N}, \ a \in \mathbb{Z}$:
$(\exists x \in \mathbb{Z} : ax \equiv 1(\mod n)) \iff \gcd(a, n) = 1$

Prop 3.25 $a, b \in \mathbb{Z}, n \in \mathbb{N}$
$(\exists x \in \mathbb{Z} : ax \equiv b \ (mod\ n)) \iff \gcd(a, n) \mid b$

Ex 3.26 (a) $\nexists x \in \mathbb{Z} : 15x \equiv 7(\mod 21)$
(b) $\exists x \in \mathbb{Z} : 15x \equiv 6(\mod 21)$

Thm 3.21 $n \in \mathbb{N}$, these are equivalent
(i) n is prime
(ii) $\mathbb{Z}/n\mathbb{Z}$ is field
(iii) $\mathbb{Z}/n\mathbb{Z}$ has no zero divisors

Ex 3.18 Finite ring $R = \mathbb{Z}/n\mathbb{Z}, \ n \in \mathbb{N}, \ x \in R$:
$\bar{x}$ unit $\iff \bar{x}$ no zero divisor
Units in R are called coprime residue classes modulo n
$(\mathbb{Z}/n\mathbb{Z})^\times$ is called multiplicative group of integers modulo n

Prop 3.17 R ring
(a) $R^\times := \{x \in R : x \text{ is unit}\}$ with $\cdot$ is (abelian) group of units
(b) $x \in R^\times \implies x$ no zero divisor
(c) R finite, then also: $x \in R$ no zero divisor $\implies x$ is unit

Def 3.19 (Commutative) ring R
with $R^\times = R \setminus \{0\}$ is called field

Ex 3.20 $\mathbb{Z}/3\mathbb{Z}$ is field, $\mathbb{Z}/4\mathbb{Z}$ and zero ring are no fields.

Def 3.13 Ring $R, x \in R : \exists y \in R \setminus \{0\} : xy = 0$,
then x is called zero divisor

Def 3.15 Ring $R, x \in R$: if $\exists y \in R : xy = 1$,
then x is called a unit

Ex 3.16 units in $\mathbb{Z}/3\mathbb{Z} : \bar{1}, \bar{2}$
units in $\mathbb{Z}/4\mathbb{Z} : \bar{1}, \bar{3}$

Ex 3.14 zero divisors in $\mathbb{Z}/3\mathbb{Z} : \bar{0}$, so has no zero divisors
zero divisors in $\mathbb{Z}/4\mathbb{Z} : \bar{0}, \bar{2}$, so has zero divisor